

SREEKAR CANGO

Senior Robotics Engineer — SLAM, Localisation & Mapping

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PROFILE

Robotics engineer with expertise in building real-time localisation, mapping, and sensor-fusion systems for autonomous robots. Specialises in SLAM, state estimation and multi-sensor fusion (LiDAR, camera, IMU, GNSS/RTK) for GNSS-denied environments, HD map generation, and production ROS2 software. Strong C++ and Python engineer with a track record of taking systems from research to deployment on real hardware.

TECHNICAL SKILLS

SLAM & Localisation: Graph-based SLAM, NDT scan matching, EKF/UKF sensor fusion (robot_localization), multi-sensor calibration (LiDAR/camera/IMU), GNSS/RTK integration, degeneracy & localisability analysis

Mapping: HD map generation, point cloud processing (PCL), map tiling & preloading, MGRS/UTM/ENU coordinate frames

Robotics & Systems: ROS2 (Jazzy/Humble), Docker, MCAP, PTP / IEEE 1588 time synchronisation, sensor drivers (Ouster, Lucid, Septentrio, NovAtel)

Languages: C++, Python · **Machine Learning:** PyTorch, TensorFlow, OpenCV, computer vision, deep learning

Tooling: Git, CI/CD (Jenkins, Bitbucket), Jira, MATLAB, Gazebo, RViz, Linux

PROFESSIONAL EXPERIENCE

Localisation Lead — Kybera (formerly Auriga and Aidrivers), London, UK

Mar 2024 – Present

- Own the end-to-end architecture of the autonomy stack, designing the localisation, mapping, and perception modules for seamless reuse across multiple vehicle types and operational domains.
- Define product requirements and decompose them into technical requirements for the localisation subsystem and the broader stack, aligning engineering delivery with operational goals.
- Led a complete redesign of the localisation and mapping framework for autonomous trucks in GNSS-denied seaport environments using multi-sensor fusion, reducing localisation error by 15% and cutting navigation downtime by 50 hours/week.
- Built the core LiDAR localisation engine plus EKF with Mahalanobis gating for GNSS in multipath seaport conditions.
- Built robust calibration methodologies for a 6-sensor perception rig (multi-LiDAR, camera, and IMU, ~200 MB/s), improving spatial alignment across the localisation, perception, and navigation stack.
- Generated seaport-scale high-definition maps via SLAM, integrating features such as traffic lights and dock zones, yielding a 20% localisation improvement and 30% better path planning.
- Deployed containerised ROS2 modules with Docker across a multi-LiDAR, multi-camera AV fleet, ensuring reliable integration across hardware platforms.
- Mentored junior engineers across the localisation and calibration stack and led adoption of agentic coding workflows (Claude Code, project-wide context standards), accelerating documentation and onboarding across the team.

Technology Associate (Officer, C10) — Citi, Chennai, India

Jul 2021 – Jul 2023

- Engineered and automated Citi Credit Risk's Loss Model Assumptions report, building a projection model over eight quarters of historical balance data to improve forecasting accuracy and reduce manual reporting effort.
- Built an incident-analyser self-service analytics tool (Power BI) consolidating Jira and ServiceNow tickets across 50+ teams, enabling anomaly detection and backlog monitoring and reducing average ticket resolution time by 10% within a month.

Software Development Engineer Intern — Visteon, Chennai, India

May 2019 – Jul 2019

- Built a real-time driver drowsiness-detection system (eye-blink, blink-rate, head-pose, and gaze tracking) in C++, Dlib, and OpenCV, optimised for edge deployment at ~20 FPS on a Raspberry Pi.

SELECTED PROJECTS

Graph-Based Optimisation and SLAM

2024

Built a graph-based SLAM system fusing compass, GPS, and landmark observations; implemented the prediction step and analysed state, covariance, and chi-squared behaviour, and explored graph pruning for efficiency.

Robotic Arm Manipulation and Object Detection

2024

Engineered a robotic arm to detect, count, and sort objects while avoiding obstacles, using computer-vision-based shape and obstacle detection and cartesian/circular motion planning, achieving a 90% success rate across varied orientations.

FinLS4: Deep Latent State Space GAN for Financial Time Series

2024

Combined latent state-space models with a GAN architecture and a novel trading loss for stock-return forecasting; validated improvements across diverse datasets (publication-backed).

Drone Trajectory and Path Planning Simulation

2023

Simulated a quadcopter using a non-linear dynamics model, implemented full-state feedback control to track desired trajectories.

Additional projects (VQ-VAE segmentation, Vision Transformer + MixUp, Poisson image editing, gesture-controlled robotic arm) on [GitHub](#).

EDUCATION

MSc Robotics and Computation — Distinction — University College London, London, UK

2023 – 2024

Thesis: “Enhanced Localisation for Autonomous Prime Movers in Seaports using Multi-Sensor Fusion”

B.E. Electronics and Communication — Distinction — Madras Institute of Technology, Chennai, India

2017 – 2021

PUBLICATIONS

- Prasad, S., Kumar, S., & Cango, S. (2024). FinLS4: Deep Latent State Space GAN for Financial Time Series. University College London — accepted at CIFAR DLRL / Vector Institute Summer School 2024.
- S. Rubin Bose, V. Sathiesh Kumar, C. Sreekar (2023). “In-situ enhanced anchor-free deep CNN framework for high-speed human-machine interaction.” *Engineering Applications of Artificial Intelligence*, Vol. 126(B), 106980.
- C. Sreekar et al. (2021). “Positioning the 5-DOF Robotic Arm using Single Stage Deep CNN Model.” ICBSII 2021, pp. 1–6. doi:10.1109/ICBSII51839.2021.9445124.

CERTIFICATIONS

[TensorFlow Developer Certificate](#) · [Deep Learning Specialization](#) · [TensorFlow Developer Specialization](#) · [HackerRank Problem Solving](#)

LEADERSHIP & AWARDS

- **Academic Representative**, MSc Robotics & Computation, UCL — represented 40 students at university level, coordinating with 30+ representatives to resolve 50+ student concerns.
- **Chairman**, MIT Robotics Association — taught weekly ML, computer vision, and embedded sessions and trained a 150+ member community to compete in national robotics challenges.
- **Giri Brothers Award** — “Best Student of the Department”; Hackathon winner (NIT Trichy), awarded £1,500.